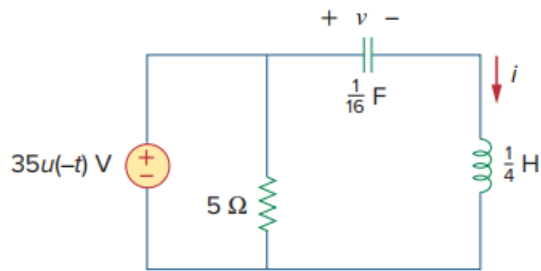


## Problem

Calculate  $i(t)$  for  $t > 0$  in the circuit of Fig. 8.82.

Figure 8.82:



## Step-by-step solution

## Step 1 of 12

Refer to Figure 8.82 in the text book.

At  $t < 0$ , the inductor acts as short circuit and the capacitor acts as open circuit. The value of the voltage source is  $35 \text{ V}$  since  $u(-t) = 1$  for  $t < 0$ .

Draw the circuit diagram at  $t < 0$ .

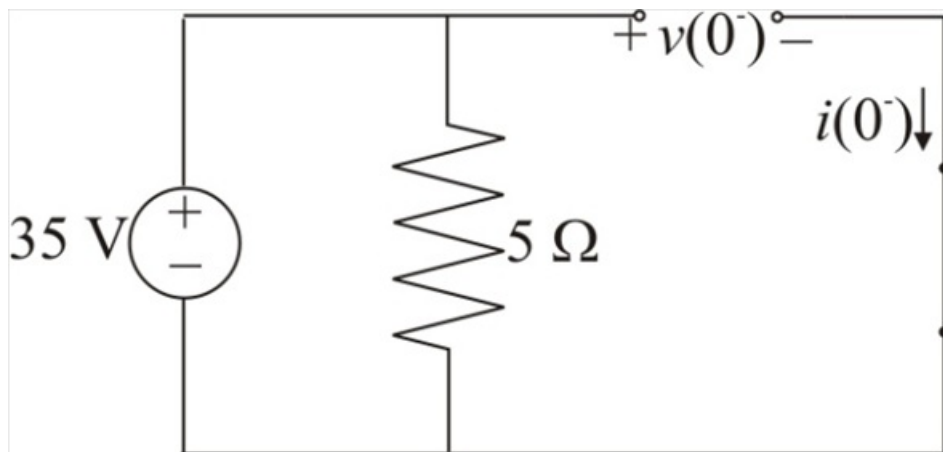


Figure 1

## Step 2 of 12

Calculate the source voltage for  $t < 0$ .

$$\begin{aligned}v_s(0^-) &= 35u(-t) \\&= (35)(1) \\&= 35 \text{ V}\end{aligned}$$

Calculate the voltage across capacitor for  $t < 0$ .

$$\begin{aligned}v(0^-) &= v_s(0^-) \\&= 35 \text{ V}\end{aligned}$$

From Figure 1, as the circuit is an open circuit.

$$i(0^-) = 0 \text{ A}$$

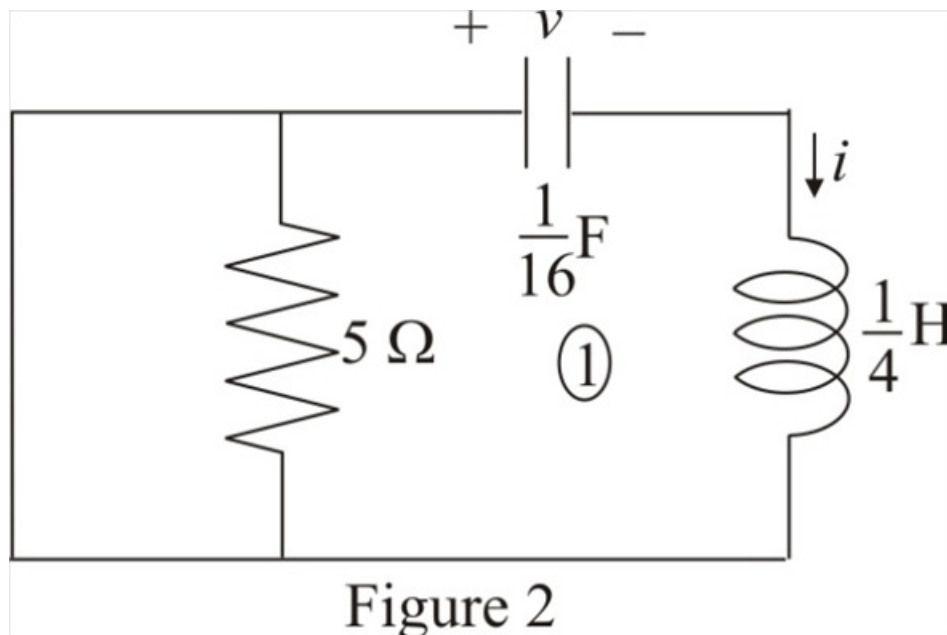
The values of the current  $i(0^-)$  and the voltage  $v(0^-)$  are 0 A and 35 V respectively.

### Step 3 of 12

Calculate the source voltage for  $t > 0$ .

$$\begin{aligned}v_s(0^+) &= 35u(-t) \\&= (35)(0) \\&= 0 \text{ V}\end{aligned}$$

Redraw the circuit diagram at  $t > 0$ .



### Step 4 of 12

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at  $t > 0$  the value of the voltage across the capacitor  $v(0^+)$  is 35 V and the value of the current  $i(0^+)$  is 0 A.

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at  $t = 0$  are  $v(0) = 35 \text{ V}$  and  $i(0) = 0 \text{ A}$ .

### Step 5 of 12

Draw the circuit for  $t = \infty$ .

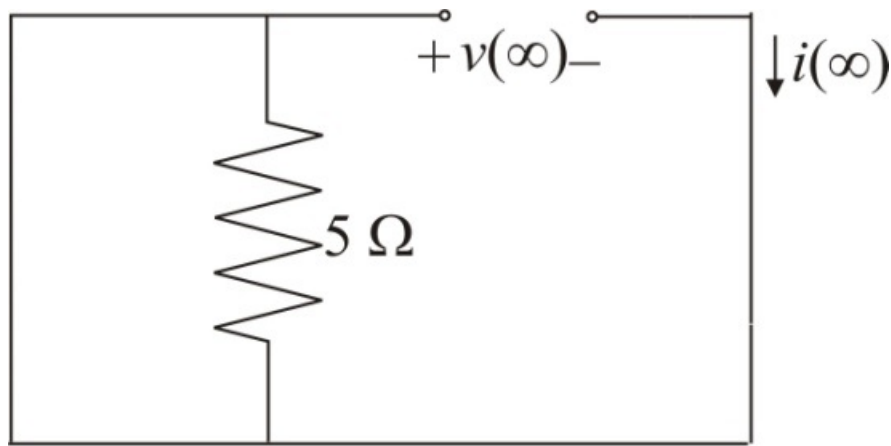


Figure 3

The value of the current at  $t = \infty$  is 0 A.

Step 6 of 12

Consider the circuit shown in Figure 3 as series  $RLC$  circuit with the value of  $R$  as 0.

Calculate neper frequency.

$$\alpha = \frac{R}{2L}$$

Substitute 0 for  $R$ .

$$\begin{aligned}\alpha &= \frac{0}{2L} \\ &= 0\end{aligned}$$

Step 7 of 12

Calculate resonant frequency  $\omega_0$ .

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Substitute  $\frac{1}{4}$  for  $L$ ,  $\frac{1}{16}$  for  $C$ .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{\left(\frac{1}{4}\right)\left(\frac{1}{16}\right)}} \\ &= 8\end{aligned}$$

Step 8 of 12

As  $\alpha < \omega_0$ , the response is underdamped.

The characteristic equation of current is

$$i(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t) \dots\dots (1)$$

Calculate damping frequency  $\omega_d$ .

$$\omega_d = \sqrt{(\omega_0^2 - \alpha^2)}$$

Substitute 0 for  $\alpha$ , 8 for  $\omega_0$ .

$$\begin{aligned}\omega_d &= \sqrt{(8^2 - 0^2)} \\ &= 8\end{aligned}$$

#### Step 9 of 12

Recall equation (1).

$$i(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$$

Substitute 8 for damping frequency  $\omega_d$ , and 0 for  $\alpha$ .

$$i(t) = (A_1 \cos 8t + A_2 \sin 8t) \dots\dots (2)$$

Substitute 0 for  $t$ .

$$\begin{aligned}i(0) &= A_1 + 0 \\ &= A_1\end{aligned}$$

Substitute 0 for  $i(0)$ .

$$A_1 = 0$$

#### Step 10 of 12

At  $t = 0$ ,

$$\begin{aligned}v_L(t) &= -v_c(t) \\ &= -35 \text{ V}\end{aligned}$$

Write the expression for voltage across inductor.

$$\begin{aligned}v_L(t) &= L \frac{di}{dt} \\ \frac{di(t)}{dt} &= \frac{1}{L} v_L(t)\end{aligned}$$

Substitute 0 for  $t$ .

$$\frac{di(0)}{dt} = \frac{1}{L} \times v_L(0)$$

Substitute  $\frac{1}{4}$  for  $L$  and  $-35$  for  $v_L(0)$ .

$$\begin{aligned}\frac{di(0)}{dt} &= \frac{1}{\left(\frac{1}{4}\right)} (-35) \\ &= (4)(-35) \\ &= -140 \text{ A/s}\end{aligned}$$

#### Step 11 of 12

Recall equation (2).

$$i(t) = (A_1 \cos 8t + A_2 \sin 8t)$$

Differentiate with respect to  $t$ .

$$\frac{di(t)}{dt} = 8A_2 \cos 8t$$

Substitute 0 for  $t$ .

$$\frac{di(0)}{dt} = 8A_2 \cos 8(0)$$

Substitute  $-140$  for  $\frac{di(0)}{dt}$ .

$$-140 = 8A_2 \cos 8(0)$$

$$8A_2 = -140$$

$$A_2 = \frac{-140}{8}$$
$$= -17.5$$

**Step 12** of 12

Substitute 0 for  $A_1$  and  $-17.5$  for  $A_2$  in  $i(t) = A_1 \cos 8t + A_2 \sin 8t$  A .

$$i(t) = (0) \cos 8t + (-17.5) \sin 8t \text{ A}$$
$$= -17.5 \sin 8t \text{ A}$$

Therefore, the current  $i(t)$  for  $t > 0$  is  $\boxed{-17.5 \sin 8t \text{ A}}$  .